ECRA: Evidence-Conditioned Reuse of LLM-Generated Annotations for Multimodal Conversational Emotion Recognition

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Abstract

Multimodal conversational emotion recognition (MERC) combines textual, acoustic, and visual cues with dialogue context to identify utterance-level emotions. However, existing methods remain susceptible to unreliable cues under modality conflicts and varying reliability. Large language models (LLMs) can provide supplementary contextual explanations, but generated judgments without supporting evidence or in conflict with observed cues may also disrupt existing predictions. To enable reliable use of generated annotations, we propose Evidence-Conditioned Reuse of LLM-Generated Annotations (ECRA). ECRA separates the decision to admit a generated annotation from the magnitude of its update to an observation-only representation. Textual and audio-visual retrieval grounds reusable annotations in labeled dialogue examples. Retrieval support and calibrated confidence guide admission; trust-scaled residual interpolation controls update magnitude. Source allocation combines observed states and admitted cognition into a proposed representation. Rejection preserves the reference representation and shared-classifier prediction within the same forward pass. Structured annotations also supply auxiliary supervision. ECRA achieves weighted F1 of 74.87 ± 0.12% and 67.51 ± 0.12% on IEMOCAP and MELD, respectively.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Human-centered computing** → *Collaborative and social computing*.

Keywords

multimodal conversational emotion recognition, large language models, evidence-conditioned reuse, cognitive caching, reliability-aware fusion

ACM Reference Format:

Anonymous Author(s). 2027. ECRA: Evidence-Conditioned Reuse of LLM-Generated Annotations for Multimodal Conversational Emotion Recognition. In *Proceedings of The ACM Web Conference 2027 (WWW '27).* ACM, New York, NY, USA, 12 pages.

1 Introduction

Multimodal conversational emotion recognition (MERC) aims to identify the emotion of each utterance in a conversation. People express emotions through language, vocal prosody, and visual behavior, and jointly analyzing these cues helps us understand speakers' emotional states and their changes. Emotional information can further complement the semantic understanding of a conversation and provide a basis for adjusting the content and style of a system's responses. These capabilities can support Web applications such as affective dialogue systems, online customer service, and multimedia interaction.

However, the emotional meaning of an utterance cannot always be determined from its explicit cues and often depends on preceding turns, speaker states, and interaction intentions. The same brief response may express different emotions in different conversational situations, and some emotional causes are not explicitly presented in the current utterance. Existing methods use context encoding and dialogue graphs to model speaker states and dependencies among utterances [2, 4, 7, 15, 27], improving conversational emotion representations. Yet, for expressions that depend on situational knowledge and implicit intentions, modeling observed cues and their relationships alone may still be insufficient to explain their emotional meaning.

In addition to contextual dependence, emotional cues from different modalities may conflict, and their relevance and reliability also vary across utterances and situations. Emotional tendencies in text may be inconsistent with vocal prosody or visual behavior, requiring the model to distinguish which cues to rely on. Existing studies model relationships among modalities through cross-modal interaction and adjust the contributions of information sources through dynamic fusion [11, 19]; reliability-aware methods further estimate prediction confidence under observation uncertainty [26]. Unreliable cues can still disturb predictions supported by other modalities, motivating explicit control over their influence.

To supplement situations and emotional causes that are not explicitly expressed in observed cues, existing studies introduce external knowledge and generated explanations. Commonsense representations and knowledge-enhanced methods provide information related to situations and emotions [25, 30]. LLMs can combine dialogue context with reference evidence to generate candidate judgments about the current utterance's emotion, providing supplementary information for multimodal emotion recognition. They can also generate coreference relations, topics, and emotional causes to enrich dialogue features [24], or supply explanatory cues through speech descriptions and cognitive reasoning [16, 31]. Generated judgments may nevertheless lack evidence support or conflict with observed cues. Moreover, connecting these judgments to a discriminative model requires control over whether they participate and how much they affect the observed representation.

Annotation reuse requires deciding whether a judgment has sufficient support to participate and how much it may change the observed representation. Screening alone cannot eliminate erroneous judgments.

We propose Evidence-Conditioned Reuse of LLM-Generated Annotations (ECRA), a controlled reuse mechanism for unreliable generated annotations. Textual and audio-visual retrieval supplies labeled dialogue examples that ground the LLM's judgments in relevant conversational evidence. Class priors derived from these judgments, affective appraisals, and modality hints form a reusable

WWW '27, Dublin, Ireland
2027.

cognitive cache. The cached fields supply supplementary prediction features and auxiliary supervision. An observation-only backbone provides a reference representation for each utterance. Evidence support and calibrated confidence determine whether its annotation is admitted. For admitted annotations, source allocation combines observed modality states with the encoded cognitive state to form a proposal; trust-scaled residual interpolation then controls how far the reference moves toward it. Rejection preserves the reference representation and shared-classifier prediction within the same forward pass.

Our main contributions are as follows:

- We propose a controlled reuse mechanism for unreliable generated annotations, linking retrieval-grounded structured inputs to prediction and auxiliary supervision.
- We separate evidence-guided admission from trust-scaled correction to control whether annotations participate and how much they update an observation-only reference.
- We test complementarity through matched-budget beneficial and harmful flips and fixed-admission injection with shared modules, supported by fusion controls and ablations.

2 Related Work

2.1 Context and Dialogue Modeling

Context modeling captures dialogue history, speaker states, and interactions. DialogueRNN tracks speaker states [15], DialogueGCN models utterance dependencies through graphs [4], and DialogueCRN performs multi-turn contextual reasoning [7]. Adversarial contrastive learning and emotion anchors improve representation discrimination [6, 34]; dynamic graphs model emotion dynamics [21], interactive hypergraphs capture complex dialogue relationships [2], and persona-aware graphs combine persona information with emotion-shift detection [27]. Other studies target unseen-speaker generalization [23], causal modeling, and representation disentanglement [17, 22]. ECRA complements these encoders with retrieval-grounded supplementary emotion annotations.

2.2 Multimodal Fusion and Reliability

Multimodal fusion combines complementary cues through interaction and contribution allocation. SDT combines intra- and inter-modal Transformers, hierarchical gated fusion, and self-distillation to strengthen observed-modality representations [14]. MultiEMO learns cross-modal correlations [19], and UniMSE jointly learns shared representations for sentiment analysis and emotion recognition [8]; semantic–emotional dual channels [32] and low-rank matching attention [20] improve fusion through feature organization and interaction modeling, respectively. DQ-Former dynamically learns modality priorities [11], and CMERC calibrates multimodal prediction confidence [26]. Related research estimates modality consistency with typicality and pseudo unimodal emotion labels [33], handles incomplete inputs through cross-modal prompting [5], and integrates temporal slicing and attention [28]. ECRA uses observed-modality fusion as a reference, separately controlling generated-annotation admission and update magnitude.

2.3 Knowledge-Enhanced and LLM-Assisted Emotion Understanding

Knowledge-enhanced methods use situational knowledge, commonsense, and emotional causes to supplement utterance cues. Emotion-oriented knowledge representations [29] and ECoK's emotional commonsense graph [30] provide external knowledge, while SKIER combines discourse relations and symbolic knowledge to construct dialogue representations [12]. MKE-IGN selects and refines textual and visual commonsense knowledge in contextual dialogue graphs [25]; joint emotion–cause modeling and evidence–cause attention connect emotions with their causes [9, 35].

MKFM integrates generated coreference relations, topics, and emotional causes [24]; speech descriptions convert acoustic expression into explanatory cues accessible to LLMs [31]; CMTD combines personality traits, cognitive-distortion analysis, and visual and speech agents, using personality traits to temper negativity bias in reasoning [16]. ECRA ties generated annotations to labeled retrieval evidence, using support and calibrated confidence to govern reuse.

ECRA evaluates evidence-conditioned annotation reuse through recognition gains, beneficial corrections, and harmful prediction flips.

3 Method

3.1 Problem Formulation and Overview

For a dialogue $D = \{u_i\}_{i=1}^{T}$ of T utterances, each u_i supplies feature vectors $x_i^m \in \mathbb{R}^{d_m}$ for modality $m \in \{t, a, v\}$ and speaker identity s_i. Here d_m is the modality-specific input dimension. Its label $y_i \in \{0, \dots, C-1\}$ belongs to C emotion classes. Superscripts t, a, v, c index text, audio, visual, and cognitive branches; $[\,;\,]$ denotes concatenation. Encoded states have hidden dimension d_h; Δ^{r-1} denotes the probability simplex over r entries. All W matrices and affine intercepts are learned parameters. Encoding and retrieval use the target and preceding turns.

Figure 1 frames annotation reuse around an observation-only reference h_i^{base}. Retrieved evidence grounds supplementary annotations; admission determines their participation, and a separate trust-scaled coefficient sets the update magnitude. Annotation fields also supply auxiliary supervision.

3.2 Observation-based Reference

The observation-side foundation uses intra-modal and cross-modal encoding, adaptive fusion, and branch self-distillation. The causal encoder uses projection $\phi_m : \mathbb{R}^{d_m} \to \mathbb{R}^{d_h}$ and learned positional and speaker embeddings $p_i^{\text{pos}}, e_i^{\text{spk}}$.

For $m, n \in \{t, a, v\}$, stack the embedded prefix as $\bar{H}_{1:i}^m$:

$$\bar{h}_i^m = \phi_m(x_i^m) + p_i^{\text{pos}} + e_i^{\text{spk}}, \tag{1}$$

$$z_i^{m \leftarrow n} = \text{Transformer}_{m \leftarrow n}(\bar{H}_{1:i}^m, \bar{H}_{1:i}^n)_i. \tag{2}$$

States $e_i^m = W_m[z_i^{m \leftarrow t}; z_i^{m \leftarrow a}; z_i^{m \leftarrow v}] + b_m \in \mathbb{R}^{d_h}$ define the reference, with encoder/fusion parameters θ:

$$\mu_i = \text{softmax}(W_b[e_i^t; e_i^a; e_i^v] + b_b),$$

$$h_i^{\text{base}} = F_\theta(x_{\le i}^{t,a,v}, s_{\le i}) = \sum_{m \in \{t,a,v\}} \mu_i^m e_i^m. \tag{3}$$

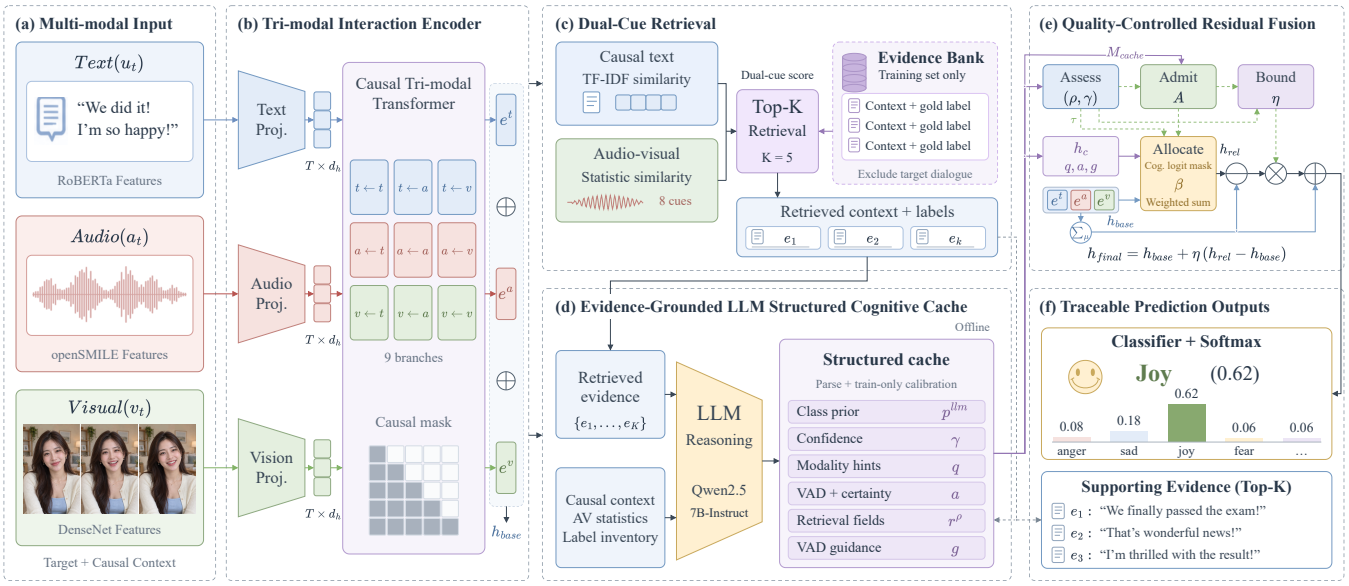


Figure 1: ECRA reuses retrieval-grounded annotations through separate admission and update-magnitude controls.

The reference reads no cache fields in the current forward pass. Branch predictions $\hat{p}_i^m \in \Delta^{C-1}$ use label supervision $\mathcal{L}_{\text{uni}}$ and self-distillation $\mathcal{L}_{\text{sd}}$ from the stopped-gradient final prediction (Appendix A.2).

3.3 Retrieval-grounded Annotation Interface

Dual-Cue Retrieval. Dual-Cue Retrieval operates independently of the encoder. For a window of w preceding turns, TF–IDF encodes the utterance texts in $C_i^w = [u_{\max(1,i-w)}, \ldots, u_i]$. Audio-visual statistics describe the current state and its shift. Cosine scores s_{text} and s_{mm} compare text and audio-visual cues. Their mixture weight $\lambda \in [0,1]$ gives score s and the top-K evidence set $\mathcal{R}_i$. The training-utterance bank $\mathcal{D}_{\text{train}}$ excludes $\mathcal{D}_{d(i)}$, where $d(i)$ indexes the target's dialogue:

$$s(u_i,u_j) = \lambda s_{\text{text}}(u_i,u_j) + (1-\lambda)s_{\text{mm}}(u_i,u_j),$$
$$\mathcal{R}_i = \text{TopK}_{u_j \in \mathcal{D}_{\text{train}} \setminus \mathcal{D}_{d(i)}}\, s(u_i,u_j). \tag{4}$$

We use $K = 5$, $w = 3$, and $\lambda = 0.7$; each item supplies its training label and causal context. The vectorizer and standardization statistics are fitted on training data; validation and test labels remain hidden during cache construction.

Cache generation. The evidence-grounded LLM builds the Structured Cognitive Cache. Qwen2.5-7B-Instruct receives causal text, audio-visual statistics, the label inventory, and retrieved examples. It returns label $\hat{y}_i^{\text{llm}}$, raw confidence γ_i^{raw}, affective appraisal, modality hints, and a rationale retained outside the discriminative input. Training-only calibration assigns out-of-fold confidence $\gamma_i \in [0,1]$ to training records and the mean of five calibrator probabilities to validation/test records. Forward computation reads this assigned value (Appendix A.1).

For a labeled cache record, the confidence-derived prior p_i^{llm} has entries for $k \in \{0,\ldots,C-1\}$. With floored confidence $\bar{\gamma}_i = \max(\gamma_i, 1/C + 10^{-6})$:

$$p_{i,k}^{\text{llm}} = \begin{cases} \bar{\gamma}_i, & k = \hat{y}_i^{\text{llm}}, \\ (1-\bar{\gamma}_i)/(C-1), & k \neq \hat{y}_i^{\text{llm}}. \end{cases} \tag{5}$$

The floor ensures a unique label mode; admission, trust, and loss weights keep γ_i unfloored. This prior uses the parsed label and confidence.

Cache fields. The appraisal $\mathbf{a}_i = [v_i, \alpha_i, d_i, \zeta_i] \in [-1,1]^4$ contains valence, arousal, dominance, and signed affective certainty. Certainty ζ_i indicates uncertainty or hesitation (< 0), indeterminacy ($= 0$), or increasing affective certainty (> 0), distinct from label confidence γ_i. Hints $q_i \in \Delta^2$ describe text, audio, and visual contributions. Support fields $r_i^{\rho} \in [0,1]^4$ are defined below; signed VAD cues $g_i \in \mathbb{R}^3$ summarize positive-valence, high-arousal negative, and low-arousal negative patterns (Appendix A.2). The cache vector is

$$c_i = [p_i^{\text{llm}}; \mathbf{a}_i; q_i; \gamma_i; r_i^{\rho}; g_i], \tag{6}$$

Its dimension is $C + 15$: 21 for IEMOCAP and 22 for MELD. Each annotation links its generated fields to the target's retrieval evidence through r_i^{ρ}. Prediction reads c_i; the same record supplies prior-alignment targets, source-weight targets, and affective guidance (Section 3.5).

3.4 Evidence-conditioned Admission and Correction

Evidence support and confidence. Evidence support summarizes retrieval consistency. Let s_i^{top} be the top combined similarity, s_i^{mean} the mean top-K similarity, and s_i^{mm} the mean audio-visual similarity. With $\tilde{s} = (s+1)/2$, let a_i^{label} be the fraction of retrieved labels matching $\hat{y}_i^{\text{llm}}$. Learned mixer logits $\theta_\rho \in \mathbb{R}^4$ combine support fields

into evidence support $\rho_i \in [0,1]$:

$$r_i^\rho = [\tilde{s}_i^{\text{top}}, \tilde{s}_i^{\text{mean}}, \tilde{s}_i^{\text{mm}}, a_i^{\text{label}}],$$
$$\rho_i = \text{softmax}(\theta_\rho)^\top r_i^\rho. \tag{7}$$

Records store fixed r_i^ρ and γ_i; the mixer recomputes support in each forward pass. The continuous trust factor τ_i is

$$\tau_i = \gamma_i(0.5 + 0.5\rho_i). \tag{8}$$

Trust modulates allocation and correction.

Annotation admission. Support and confidence are compared with thresholds δ_q, δ_c using indicator function $\mathbb{I}$. The mask $M_i^{\text{cache}} \in \{0,1\}$ indicates whether a labeled cache record is available; the quality-pass mask M_i^{llm} determines binary admission A_i:

$$M_i^{\text{llm}} = \mathbb{I}(\rho_i \geq \delta_q)\mathbb{I}(\gamma_i \geq \delta_c),$$
$$A_i = M_i^{\text{cache}} M_i^{\text{llm}}. \tag{9}$$

Admission indicators are non-differentiable. Admitted records update the shared support mixer through continuous paths. A rejected record with a valid cache and $\gamma_i \geq \delta_c$ may re-enter when subsequent shared-mixer updates raise its support to $\rho_i \geq \delta_q$.

Proposed representation. Source allocation weights the observed modalities and admitted cognition. A learned projection maps cache fields to $h_i^c = \psi(c_i) \in \mathbb{R}^{d_h}$. For unavailable records, the cache input is zero-filled and $M_i^{\text{cache}} = 0$. Modality states and cache fields determine logits $o_i \in \mathbb{R}^4$ and bias $b_i \in \mathbb{R}^4$:

$$o_i = W_r[e_i^t; e_i^a; e_i^v; h_i^c] + b_r,$$
$$b_i = W_h[q_i; \tau_i] + W_{\text{vad}}[\mathbf{a}_i; g_i], \tag{10}$$

Masked logits $\tilde{o}_i$ set the cognitive entry to o_i^c for $A_i = 1$ and $-\infty$ otherwise, leaving observed entries unchanged. Source weights $\beta_i \in \Delta^3$ form the proposed state $h_i^{\text{rel}} \in \mathbb{R}^{d_h}$:

$$\beta_i = \text{softmax}(\tilde{o}_i + b_i),$$
$$h_i^{\text{rel}} = Q_\varphi(e_i^t, e_i^a, e_i^v, c_i; A_i)$$
$$= \sum_{m \in \{t,a,v\}} \beta_i^m e_i^m + \beta_i^c h_i^c. \tag{11}$$

Here Q_φ comprises the cache projection, support mixing, and four-source allocation, with parameters φ. The mask sets $\beta_i^c = 0$ upon rejection.

Update magnitude and fallback. Let $\Delta h_i = h_i^{\text{rel}} - h_i^{\text{base}}$ and factor the correction coefficient as $\eta_i = A_i\tau_i\xi_i$. With logistic σ, learned scalars θ_η, b_η, and scalar map W_η:

$$h_i^{\text{final}} = h_i^{\text{base}} + \eta_i \Delta h_i = h_i^{\text{base}} + A_i\tau_i\xi_i\Delta h_i, \tag{12}$$
$$\xi_i = \sigma(\theta_\eta)\left[0.5 + 0.5\sigma(W_\eta[\gamma_i; \rho_i] + b_\eta)\right]. \tag{13}$$

A_i controls participation; for admitted records, $\tau_i\xi_i$ sets the step toward the four-source proposal. Since $0 \leq \xi_i \leq 1$, $0 \leq \eta_i \leq A_i\tau_i \leq 1$: the update is a convex interpolation with $\|h_i^{\text{final}} - h_i^{\text{base}}\|_2 = \eta_i\|\Delta h_i\|_2$. Observed-state coefficients are $(1-\eta_i)\mu_i^m + \eta_i\beta_i^m$, and cognition's coefficient is $\eta_i\beta_i^c$.

Rejection recovers the observation reference:

$$A_i = 0 \implies \beta_i^c = 0, \quad \eta_i = 0, \quad h_i^{\text{final}} = h_i^{\text{base}}. \tag{14}$$

Table 1: Dataset statistics.

Dataset	C	Conversations			Utterances		
		Train+Val	Test	Total	Train+Val	Test	Total
IEMOCAP	6	120	31	151	5,810	1,623	7,433
MELD	7	1,153	280	1,433	11,098	2,610	13,708

The shared classifier produces $\hat{p}_i = \text{softmax}(W_o h_i^{\text{final}} + b_o)$ and $\hat{y}_i = \arg\max_k \hat{p}_{i,k}$. Applied to h_i^{base}, it yields $\hat{p}_i^{\text{base}}$. Equation 14 therefore gives $\hat{p}_i = \hat{p}_i^{\text{base}}$ in the same forward pass when $A_i = 0$.

3.5 Annotation-derived Supervision

Judgment targets. LLM-Driven Alignment, $\mathcal{L}_{\text{llm}}$, aligns the final prediction with class priors from admitted cache records, weighted by admission, support, and confidence.

Source-weight supervision. The reliability loss $\mathcal{L}_{\text{rel}}$ aligns β_i with a four-source target β_i^* derived from hints, VAD cues, support, and confidence. The cognitive share is $\pi_i^c = \gamma_i\rho_i\chi_i$, with appraisal strength $\chi_i \in [0,1]$; normalized hints and appraisal-derived cues allocate the rest.

Auxiliary sample weights and their normalizers in $\mathcal{L}_{\text{llm}}$ and $\mathcal{L}_{\text{rel}}$, together with the target β_i^*, are recomputed each forward pass but detached during back-propagation. Gradients through $\hat{p}_i$ and β_i remain enabled, updating the support mixer through continuous allocation and correction paths.

Affective geometry. Affective Guidance, $\mathcal{L}_{\text{aff}}$, uses VAD appraisals for weighted same-class attraction and a margin for training-derived confusable label pairs. Cache-dependent objectives use admitted records.

Supervised and cognitive loss groups $\mathcal{L}_{\text{sup}}$ and $\mathcal{L}_{\text{cog}}$ form the total loss $\mathcal{L}$; each nonnegative $\lambda_\cdot$ weights its corresponding loss:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{cog}} + \lambda_{\text{aff}}\mathcal{L}_{\text{aff}},$$
$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{uni}}\mathcal{L}_{\text{uni}} + \lambda_{\text{sd}}\mathcal{L}_{\text{sd}},$$
$$\mathcal{L}_{\text{cog}} = \lambda_{\text{llm}}\mathcal{L}_{\text{llm}} + \lambda_{\text{rel}}\mathcal{L}_{\text{rel}}. \tag{15}$$

Appendix A.2 gives the cues, full losses, targets, and pair definitions.

4 Experiments

4.1 Experimental Setup

Datasets and Splits. IEMOCAP [1] contains scripted and improvised dyadic conversations between actors, with happy, sad, neutral, angry, excited, and frustrated labels. Validation uses a fixed-seed 10% of 120 development dialogues. MELD [18] contains multi-party conversations from *Friends*, with neutral, surprise, fear, sadness, joy, disgust, and anger labels; supplied partitions are retained (Table 1).

Baselines. We select representative methods from the following three categories for comparison:

1. Context and Dialogue Modeling: DialogueRNN [15] tracks speaker states; DialogueGCN [4] models dialogue dependencies through graph convolution; DialogueCRN [7] performs multi-turn contextual reasoning; SACL [6] uses adversarial contrastive learning; EACL [34] introduces emotion anchors; DGODE [21] models emotion dynamics using graph neural ODEs; DIB-HGCN [2]

Table 2: Published baseline results.

Method	IEMOCAP									MELD									
	Class-wise F1						Overall			Class-wise F1							Overall		
	Happy	Sad	Neut.	Ang.	Exc.	Frus.	Acc.	W-F1	M-F1	Neut.	Surp.	Fear	Sadn.	Joy	Disg.	Ang.	Acc.	W-F1	M-F1
DialogueRNN	33.83	69.83	57.76	62.50	64.45	59.46	59.33	59.89	57.97	75.76	48.23	2.77	17.93	53.04	2.58	42.21	55.27	56.93	34.65
DialogueGCN	42.75	84.54	63.54	64.19	63.08	66.99	65.25	64.18	64.18	77.82	54.11	2.31	36.43	55.07	2.12	47.22	60.96	60.98	39.30
DialogueCRN	51.59	74.54	62.38	67.25	73.96	59.97	65.31	65.34	64.95	76.13	46.55	0.00	11.43	49.47	0.00	44.92	59.66	56.76	32.64
SACL	56.91	84.78	70.00	64.09	69.70	65.02	69.08	69.22	68.42	80.17	58.77	26.23	41.34	64.98	31.47	52.35	67.51	66.45	50.76
EACL	51.29	81.80	73.32	67.54	71.27	67.76	–	70.41	68.83	80.44	60.48	23.54	42.41	65.22	33.86	54.01	–	67.12	51.42
DGODE	71.80	71.00	74.90	55.70	78.60	75.20	–	72.80	71.20	82.60	60.90	5.10	45.50	63.40	10.60	54.00	–	67.20	46.01
DIB-HGCN	61.36	79.05	71.45	70.56	81.88	66.85	72.15	72.45	71.86	80.17	60.35	6.78	44.39	64.81	23.26	54.75	67.82	66.57	47.79
PCGNet	49.83	82.7	71.62	69.14	76.08	70.98	71.72	71.77	70.06	80.25	61.02	25.88	41.48	64.65	25.24	56.09	67.85	67.02	50.66
SGF	59.75	78.72	71.39	70.83	80.13	65.52	–	71.64	71.06	80.40	61.01	18.18	44.07	64.07	29.47	54.43	–	66.95	50.23
CMERC	60.73	81.89	71.65	69.51	77.45	67.02	–	71.98	71.38	80.18	60.42	24.69	40.48	65.30	32.31	54.16	–	66.85	51.08
DQ-Former	61.67	69.61	71.87	77.39	80.95	67.71	71.68	71.76	71.53	78.31	59.13	21.90	39.93	60.84	26.54	52.82	64.88	64.70	48.50
SEDC	62.78	84.49	71.52	65.71	73.63	68.09	71.60	71.68	71.04	80.09	58.11	23.38	40.62	65.22	28.30	52.16	67.43	66.16	49.70
LMAM	74.31	85.47	67.03	65.26	80.14	68.38	73.67	73.01	73.43	74.28	64.74	2.14	42.35	62.52	7.05	60.26	64.93	63.88	44.76
SDT	66.19	81.84	74.62	69.73	80.17	68.68	73.95	74.08	73.54	80.19	59.07	17.88	43.69	64.29	28.78	54.33	67.55	66.60	49.75
MKE-IGN	53.87	82.86	72.07	71.30	75.78	68.84	72.03	71.93	70.79	80.00	59.82	17.39	40.12	64.01	31.30	56.08	67.78	66.56	49.82
ECERC	60.86	79.28	71.95	66.27	78.29	68.25	71.60	71.78	70.82	79.80	58.98	26.12	40.95	64.95	31.43	53.89	67.32	66.46	50.87
ECRA	67.69	83.26	74.97	70.45	79.72	70.78	**74.86**	**74.99**	**74.48**	80.35	61.79	25.00	40.91	65.23	31.86	58.18	**68.58**	**67.62**	**51.90**

constructs monologic and dialogic sub-hypergraphs with different windows; PCGNet [27] combines persona information with emotion-shift detection; SGF [23] handles unseen speakers through speaker prototypes and contrastive learning.

2. Multimodal Fusion and Reliability: CMERC [26] calibrates multimodal prediction confidence; DQ-Former [11] dynamically learns modality priorities; SEDC [32] separates semantic and emotional channels; LMAM [20] uses low-rank matching attention; SDT [14] strengthens modality representations through self-distillation; UniMSE [8] jointly models sentiment analysis and emotion recognition through multimodal contrastive learning.

3. Knowledge-, Cause-, and LLM-Enhanced Methods: MKE-IGN [25] selects and refines textual and visual commonsense knowledge; ECERC [35] relates multimodal evidence to contextual emotional causes; MKFM+ACS [24] integrates emotion-related knowledge generated by ChatGPT.

Comparison Protocol. Table 2 summarizes publicly reported baseline results under their reported configurations and ECRA's designated reference-run result. Because features and training configurations differ across studies, Table 3 further provides matched reproductions of selected baselines. Reproductions preserve original architectures and the same partitions, reusing common input features where compatible. Method-specific inputs and preprocessing follow original settings (Appendix B.1). Trained discriminative variants report five-training-run mean ± sample SD; direct predictors reuse precomputed outputs. Single-run performance comparisons use the designated reference run. Fixed-checkpoint counts and perturbation variability are labeled separately. For Table 3, two-sided Welch t-tests compare the five run-level test W-F1 scores of each baseline with ECRA, using training runs as the statistical units. Holm correction covers the nine baseline–ECRA comparisons separately within each dataset; all adjusted p-values are below 0.001.

Evaluation Metrics. We use weighted F1 (W-F1) as the primary metric and also report accuracy (ACC), class-wise F1, and Macro-F1 (M-F1). W-F1 weights class-wise F1 by the number of samples in each class, whereas M-F1 gives each class equal weight, complementing the evaluation of recognition performance under class imbalance. Table 2 computes M-F1 from the displayed class-wise scores. Unreported metrics are denoted by –. Accuracy and F1 scores are reported as percentages.

Implementation Details. RoBERTa-large [13], openSMILE [3], and DenseNet-121 [10] provide textual, acoustic, and visual features with dimensions 1024, 1,582/300 (IEMOCAP/MELD), and 342, respectively. ECRA uses a hidden dimension of 1024, dropout of 0.5, eight attention heads, and 100/50 training epochs on IEMOCAP/MELD. Adam uses $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$ and weight decay 10^{-5} without gradient clipping. The batch size and learning rate are 16 and 10^{-4} for IEMOCAP, and 8 and 5×10^{-6} for MELD; learning rates remain constant. Hyperparameters and model checkpoints are selected by validation W-F1, and the test set is used for final evaluation.

Cache Generation and Calibration. Qwen2.5-7B-Instruct generates caches offline through greedy decoding. Dialogue-level five-fold calibration uses training labels only: training forwards and diagnostics use out-of-fold confidence; validation/test use the mean of five calibrator probabilities. Both datasets use validation-selected thresholds $\delta_q = 0.60$ and $\delta_c = 0.85$.

4.2 Overall Results

In Table 2, ECRA's reference-run ACC, W-F1, and M-F1 lead the listed baselines on both datasets. IEMOCAP W-F1 is 74.99%, exceeding SDT's 74.08% by 0.91 percentage points; MELD W-F1 is 67.62%, exceeding DGODE by 0.42 points.

Table 3: Matched baseline reproductions.

Method	ACC	W-F1	M-F1
(a) IEMOCAP			
DialogueCRN	65.24 ± 0.15	65.29 ± 0.17	64.72 ± 0.12
DIB-HGCN	72.08 ± 0.16	72.37 ± 0.18	71.82 ± 0.10
PCGNet	71.51 ± 0.25	71.67 ± 0.28	70.60 ± 0.22
DQ-Former	71.47 ± 0.23	71.69 ± 0.23	71.18 ± 0.30
SEDC	71.44 ± 0.18	71.63 ± 0.15	70.92 ± 0.12
SDT	73.85 ± 0.06	73.98 ± 0.09	73.40 ± 0.10
UniMSE	70.44 ± 0.14	70.59 ± 0.16	69.69 ± 0.20
ECERC	71.60 ± 0.24	71.78 ± 0.18	71.04 ± 0.34
MKFM+ACS	68.64 ± 0.19	68.80 ± 0.25	67.35 ± 0.30
ECRA	**74.71 ± 0.13**	**74.87 ± 0.12**	**74.34 ± 0.19**
(b) MELD			
DialogueCRN	62.47 ± 0.22	55.78 ± 0.24	31.72 ± 0.19
DIB-HGCN	67.77 ± 0.12	66.52 ± 0.09	47.66 ± 0.12
PCGNet	67.80 ± 0.14	66.97 ± 0.15	50.63 ± 0.12
DQ-Former	66.46 ± 0.58	64.89 ± 0.22	43.43 ± 1.42
SEDC	67.16 ± 0.21	66.13 ± 0.12	49.47 ± 0.27
SDT	67.45 ± 0.07	66.50 ± 0.07	49.09 ± 0.71
UniMSE	66.99 ± 0.09	65.43 ± 0.10	43.24 ± 0.22
ECERC	67.32 ± 0.19	66.46 ± 0.27	49.35 ± 0.92
MKFM+ACS	66.95 ± 0.14	65.22 ± 0.14	42.69 ± 0.18
ECRA	**68.34 ± 0.15**	**67.51 ± 0.12**	**51.70 ± 0.21**

Table 4: Component ablations.

Setting	IEMOCAP		MELD	
	ACC	W-F1	ACC	W-F1
Base / w/o cache	70.52 ± 0.26	70.76 ± 0.11	65.89 ± 0.36	64.43 ± 0.08
Shuffled cache	70.60 ± 0.31	70.85 ± 0.20	65.99 ± 0.40	64.04 ± 0.26
w/o admission	71.66 ± 0.14	71.98 ± 0.15	65.96 ± 0.21	64.83 ± 0.14
w/o residual	72.43 ± 0.18	72.73 ± 0.13	65.92 ± 0.07	64.80 ± 0.17
w/o ρ	72.55 ± 0.26	72.81 ± 0.17	67.14 ± 0.13	65.86 ± 0.10
w/o γ	72.69 ± 0.14	72.96 ± 0.16	67.00 ± 0.45	65.68 ± 0.14
w/o weight loss	72.63 ± 0.11	72.79 ± 0.18	67.48 ± 0.19	65.95 ± 0.14
w/o LLM alignment	72.78 ± 0.08	73.05 ± 0.07	67.17 ± 0.33	65.78 ± 0.10
w/o affective loss	73.34 ± 0.19	73.55 ± 0.12	67.69 ± 0.10	66.33 ± 0.21
Full ECRA	**74.71 ± 0.13**	**74.87 ± 0.12**	**68.34 ± 0.15**	**67.51 ± 0.12**

The matched reproductions in Table 3 maintain this advantage. ECRA's W-F1 is 74.87 ± 0.12% and 67.51 ± 0.12% on IEMOCAP and MELD, respectively, improving over DIB-HGCN by 2.50 and 0.99 percentage points; ACC and M-F1 also lead. DIB-HGCN models monologic and dialogic relations, while ECRA supplements contextual multimodal representations with retrieval-grounded emotion judgments from the cognitive cache.

Compared with DQ-Former, which dynamically learns modality priorities, ECRA improves W-F1 by 3.18 and 2.62 percentage points on the two datasets, respectively. Section 4.4 compares integration systems trained with a shared cache.

On IEMOCAP and MELD, ECRA improves W-F1 over ECERC by 3.09 and 1.05 points, and over MKFM+ACS by 6.07 and 2.29 points, respectively. Table 4 tests judgment screening and correction control through targeted ablations.

Table 5: Integration controls.

Setting	IEMOCAP		MELD	
	ACC	W-F1	ACC	W-F1
Base	70.52 ± 0.26	70.76 ± 0.11	65.89 ± 0.36	64.43 ± 0.08
ECRA-base	71.18 ± 0.15	71.36 ± 0.16	66.28 ± 0.16	65.40 ± 0.25
ECRA-base + prob.	71.81 ± 0.16	71.98 ± 0.16	66.85 ± 0.14	65.99 ± 0.16
ECRA-base + logit	71.94 ± 0.17	72.10 ± 0.16	67.01 ± 0.12	66.14 ± 0.17
RFB	72.18 ± 0.17	72.40 ± 0.15	67.41 ± 0.24	65.93 ± 0.19
Non-LLM cache	72.54 ± 0.26	72.90 ± 0.22	67.43 ± 0.18	66.15 ± 0.20
E2E Cache Gate	73.66 ± 0.20	73.82 ± 0.16	67.84 ± 0.04	66.87 ± 0.09
E2E-ηGate	71.97 ± 0.26	72.15 ± 0.31	66.98 ± 0.13	66.03 ± 0.16
Retrieval-only	58.78 ± 0.00	58.75 ± 0.00	61.34 ± 0.00	54.04 ± 0.00
LLM-only	63.40 ± 0.00	63.58 ± 0.00	63.14 ± 0.00	59.59 ± 0.00
RAG-LLM	68.33 ± 0.00	68.54 ± 0.00	66.74 ± 0.00	65.25 ± 0.00
ECRA-text	72.86 ± 0.14	73.12 ± 0.11	67.68 ± 0.11	66.57 ± 0.14
ECRA-AV	71.37 ± 0.26	71.69 ± 0.17	67.05 ± 0.35	65.64 ± 0.20
ECRA	**74.71 ± 0.13**	**74.87 ± 0.12**	**68.34 ± 0.15**	**67.51 ± 0.12**

4.3 Ablation Study

Table 4 tests whether annotations must match their targets, whether judgments should be screened, and whether admitted proposals should adjust or replace the reference. Other training settings remain unchanged.

Removing the cache and its dependent paths and losses yields Base (Table 5), lowering W-F1 by 4.11/3.08 points. Shuffled cache retains complete records but permutes their target correspondence, reducing W-F1 to 70.85% and 64.04%. Matching cache dimensions and overall distribution is insufficient when records no longer match their targets.

Allowing all valid caches into fusion (w/o admission) lowers W-F1 by 2.89 and 2.68 percentage points on the two datasets, respectively. Directly classifying the fused state (w/o residual) lowers W-F1 by 2.14 and 2.71 points despite retaining admission and source weighting. Screening judgments and constraining the subsequent update both improve aggregate recognition performance.

Removing evidence support ρ or calibrated confidence γ and their associated control terms lowers IEMOCAP W-F1 to 72.81% and 72.96%, respectively, and MELD W-F1 to 65.86% and 65.68%. These recognition ablations assess confidence-conditioned control paths; calibration reliability is evaluated separately in Section 4.5.

Removing source-weight supervision lowers W-F1 by 2.08/1.56 points; allocation remains trainable through other objectives. LLM-alignment and affective-loss ablations lose 1.82/1.73 and 1.32/1.18 points, respectively. Table 4's loss-only variants retain cache fields, whereas the prior- and VAD-path ablations in Appendix B.1 remove both inputs and associated supervision.

To assess generated fields beyond fixed class templates, we retain the original LLM labels, confidence, class priors, and retrieval evidence, replacing appraisal (including ζ) and modality hints with class templates indexed by the LLM-predicted label (Appendix A.1). Replacement covers training, validation, and test caches; derived cues and auxiliary targets are recomputed, with all losses retained. Both downstream variants train independently from paired initializations under the main training and validation-selection protocol. Across five paired runs, Full ECRA achieves mean W-F1 of 74.87/67.51% on IEMOCAP/MELD, versus 73.45/66.65% for the class-template variant. Generated fields improve W-F1 by 1.42/0.86 percentage points, respectively.

Table 6: Cache selectivity and calibration.

Diagnostic	IEMOCAP	MELD
Available caches N	1,623	2,610
Accepted correct / accepted	500 / 624	691 / 884
Rejected correct / rejected	609 / 999	1,051 / 1,726
Admission (%)	38.45	33.87
All-cache label ACC (%)	68.33	66.74
Accepted label ACC (%)	80.13	78.17
Rejected label ACC (%)	60.96	60.89
ECE (%)	23.0 → 8.0	25.0 → 9.0
Brier	0.27 → 0.22	0.28 → 0.23
AUROC ($\rho/\gamma/\tau$)	0.70/0.72/0.76	0.66/0.68/0.72

4.4 Annotation Sources and Fusion Controls

Table 5 compares integration schemes using the same cache. Base + prob. and the ECRA-base + prob./logit variants train independently from scratch with fusion active during training and evaluation (Appendix B.1). ECRA exceeds ECRA-base + prob./logit by 2.89/1.52 and 2.77/1.37 W-F1 points on IEMOCAP/MELD in these system-level comparisons. ECRA-base retains cache-trained representations but disables the update at evaluation. Native fusion adds 3.51/2.11 points over this backbone-only readout in the same checkpoint.

RFB retains Base's modality weights and adds a cache-free width-12 GELU residual fusion block with 36,928 parameters and initial correction coefficient 0.05, sharing Base's objectives and initialization (Appendix B.1). ECRA improves W-F1 by 2.47/1.58 points over RFB and by 1.97/1.36 over the retrieval-derived Non-LLM cache.

Retrieval-only predicts from retrieved labels, LLM-only omits examples, and RAG-LLM directly predicts from evidence-grounded LLM judgments. Retrieved examples raise LLM-only W-F1 from 63.58/59.59 to 68.54/65.25. Dual-cue retrieval within ECRA improves W-F1 by 1.75/0.94 points over text-only retrieval and also exceeds AV-only retrieval.

E2E-ηGate retains ECRA's four-source proposal and direct $\mathcal{L}_{\text{rel}}$ supervision, replacing the trust-bounded outer coefficient with an MLP gate. Quality information remains in admission and allocation. It trains separately with its own support mixer; ECRA gains 2.72/1.48 W-F1 points in this system comparison. The earlier E2E Cache Gate changes both residual direction and coefficient (Appendix B.1); Section 4.6 adds a shared-proposal outer-gate comparison.

4.5 Admission and Correction Diagnostics

Table 6 evaluates cached LLM labels for test utterances. The admitted subset has label accuracy 80.13/78.17%, versus 68.33/66.74% for all valid caches, retaining 624/884 records. Of these, 124/193 remain incorrect, motivating correction control after quality screening.

Table 6 evaluates the same valid test caches. Training-only temperature calibration improves confidence reliability: ECE falls from 23.0/25.0% to 8.0/9.0% and binary Brier from 0.27/0.28 to 0.22/0.23 on IEMOCAP/MELD. The complete confidence-conditioned mechanism improves recognition (Table 4). For correctness ranking, τ has AUROC 0.76/0.72, versus 0.72/0.68 for γ and 0.70/0.66 for ρ.

In the main-run diagnostics, B counts wrong-to-correct flips and H correct-to-wrong flips. ECRA's B/H counts are 84/20 on IEMOCAP and 93/27 on MELD, relative to its same-forward backbone. Net corrections are 64/66, versus 16/21 for E2E Cache Gate.

Table 7: Matched-budget prediction flips.

Admission	Correction	IEMOCAP		MELD	
		$B\uparrow$	$H\downarrow$	$B\uparrow$	$H\downarrow$
Conf. only	MLP	61.4 ± 2.61	26.2 ± 2.39	72.8 ± 1.79	33.2 ± 1.64
Conf. + evidence	MLP	65.6 ± 2.41	21.6 ± 2.70	77.8 ± 0.84	29.4 ± 2.07
Conf. only	Trust-scaled	62.2 ± 1.48	19.0 ± 2.74	76.4 ± 2.19	27.0 ± 4.06
Conf. + evidence	Trust-scaled	**68.4 ± 2.51**	**15.4 ± 0.89**	**81.6 ± 0.89**	**23.2 ± 3.11**

Conf. denotes confidence. Budgets are 487/783 on IEMOCAP/MELD; correction rules share admitted IDs within each policy.

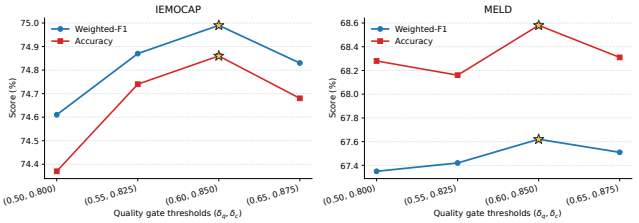


Figure 2: Effect of admission thresholds. Configurations are retrained and selected by validation W-F1.

Reference-run ACC remains 74.86/68.58%; Appendix B.2 gives the full counts and conditional-harm statistics.

For high-confidence incorrect cache labels, ECRA's conditional harmful-flip rates are 6/82 (7.32%) and 9/121 (7.44%), versus 23.26/20.00% for E2E Cache Gate. Each denominator counts the corresponding system's correct backbone predictions within this subset.

Table 7 reports mean ± sample SD across five reference checkpoints. Within each checkpoint, all four configurations share a frozen observation backbone, cache encoder, source allocation/proposal module, evidence mixer, and classifier, with fixed annotation caches and quality scores. The MLP uses E2E-ηGate's outer architecture, initialized afresh. Both correction rules minimize classification cross-entropy on the same valid training caches, updating only their outer-gate parameters. Each fitted gate is selected by mean validation W-F1 across both admission policies and reused for both. Admission policies may select different records; within each, correction rules share admitted IDs. Appendix B.2 gives the selection rules and Net statistics.

At matched budgets, the controls have complementary effects: evidence guidance and trust scaling each increase mean B and reduce mean H under either setting of the other policy. The full combination reduces mean H by 41.2/30.1% relative to the first row (ratios of means). Mean net corrections ($B - H$) rise from 35.2/39.6 to 53.0/58.4, gains of 17.8/18.8 on IEMOCAP/MELD. Mean harmful flips remain 15.4/23.2.

Figure 2 shows that the validation-selected thresholds give the highest test W-F1 among the compared configurations; relaxing or tightening them reduces performance. Appendix B.2 reports admission stability; soft and straight-through gates (Appendix B.1) trail ECRA by 2.09/1.16 and 2.03/1.09 W-F1 points.

Table 8: Fixed-admission cache injection.

Dataset	Gate	Clean W-F1 ↑	$\Delta_{0.1}$ ↓	$\Delta_{0.3}$ ↓	$\Delta_{0.5}$ ↓
IEMOCAP	MLP	73.74	0.8	2.4	4.1
	Trust-scaled	74.29	0.7	1.8	3.0
MELD	MLP	67.08	0.2	1.4	2.7
	Trust-scaled	67.46	0.3	1.0	2.0

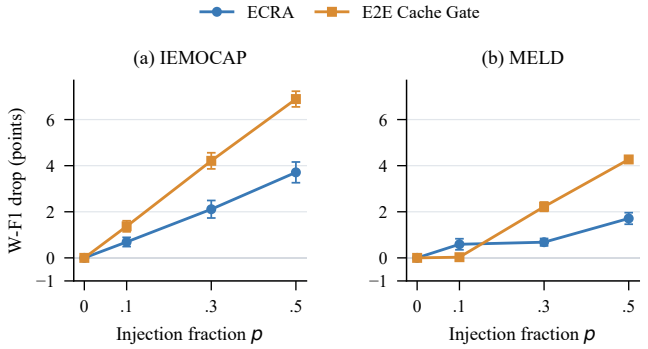


Figure 3: System-level wrong-label cache injection. Drops use each fixed checkpoint's clean W-F1; error bars show sample SD over five matched perturbations, conditional on the checkpoints.

4.6 Robustness under Perturbations

Cache injection. We keep observed content clean and inject incorrect cached labels at fractions $p \in \{0, 0.1, 0.3, 0.5\}$. Figure 3 compares ECRA and the earlier E2E Cache Gate as complete systems: they differ in residual direction and coefficient, and compute their own admission. They share perturbed records and incorrect labels, with post-calibration confidence set to 0.95. We rebuild class priors and retrieved-label agreement and recompute support with each frozen mixer; other fields stay fixed. Each uses one validation-selected checkpoint across all fractions and its own clean baseline. Five matched perturbation seeds measure variability conditional on these checkpoints (Appendix B.3).

At $p = 0.3$, ECRA loses 2.11/0.68 W-F1 points versus 4.21/2.22 for E2E Cache Gate; at $p = 0.5$, the drops are 3.71/1.71 versus 6.89/4.27. At $p = 0.1$, its MELD drop is larger (0.59 versus 0.03).

Fixed-admission comparison. Following Table 7's shared frozen-module setup, both outer gates are fitted on clean training caches with matched initial mean update norms and the same validation criterion. The MLP uses E2E-ηGate's outer architecture. After selection, all parameters stay fixed across injection conditions. We fix clean admission sets at 487/783 examples on IEMOCAP/MELD and set injected-label confidence to 0.95. Class priors, retrieved-label agreement, ρ_i, τ_i, and proposals are recomputed with unchanged parameters; both gates share perturbed caches, proposals, and admitted IDs. Table 8 reports clean W-F1 (%) and decreases Δ_p (points) from each gate's own clean score. Trust-scaled interpolation has higher clean W-F1 and smaller drops at $p = 0.3$ and $p = 0.5$. At $p = 0.1$, its MELD drop is slightly larger (0.3 versus 0.2 points).

Table 9: LLM generator comparison.

Generator	IEMOCAP		MELD	
	ACC	W-F1	ACC	W-F1
Qwen2.5-1.5B	73.07	73.26	67.89	67.10
Qwen2.5-3B	73.75	74.00	68.05	67.27
Qwen2.5-7B	74.86	74.99	68.58	67.62
Qwen2.5-14B	74.68	74.83	68.31	67.46
Llama-3.1-8B	73.81	73.97	67.85	66.82
Mistral-7B-v0.3	73.94	74.08	67.97	67.11
InternLM3-8B	74.12	74.27	68.12	67.25

Table 7 tests complementarity at matched budgets; Table 8 evaluates correction scaling with shared admitted IDs and proposals. These controls target the reuse mechanism more directly than component removal.

Input corruption. We randomly zero selected utterance-level pooled modality features and replace corresponding text in retrieval queries and prompts with `[MASK]`. Each condition reruns retrieval, recomputes audio-visual statistics, and regenerates annotation caches; both methods retain the same auxiliary supervision. At 10% input corruption, ECRA loses 2.13/1.68 W-F1 points on IEMOCAP/MELD, versus 5.22/4.17 for E2E Cache Gate. This evaluates robustness when observed cues, retrieval evidence, and generated judgments change together.

4.7 Generator Choice and Computational Cost

Generator comparison. Table 9 compares seven instruction-tuned generators and their chat serialization under shared prompts, greedy decoding, calibration, and downstream protocols. Qwen2.5-7B was fixed before testing. W-F1 spans 1.73/0.80 points on IEMOCAP/MELD, indicating limited dependence on the tested generator.

Gains over Base persist across the tested families, which populate the same class-prior, appraisal, and modality-hint fields for a common downstream interface.

Computational cost. With existing caches, Base/ECRA use 79.69M/79.73M parameters and 2.61/4.47 seconds per epoch on IEMOCAP, and 78.39M/78.43M and 14.97/22.30 seconds on MELD, excluding the LLM and feature extractors. ECRA adds about 0.05% parameters; epoch time rises by 71.26%/48.96%. At 100/50 epochs, estimated full-budget times are 261/447 and 748.5/1,115 seconds.

Caches occupy 41.57/66.50 decimal MB. Batch-16 construction takes 1.68–2.70/2.80–3.73 hours across logged settings, reported as amortized bulk cost. Appendix A.1 gives wall/loop definitions and CPU timings.

5 Conclusion

ECRA provides a controlled reuse mechanism for unreliable generated annotations, separating admission from correction magnitude. Retrieved evidence and calibrated confidence guide participation; trust-scaled interpolation regulates updates to the observation-only reference. Matched-budget and fixed-admission diagnostics with shared modules support their complementary roles on IEMOCAP and MELD; main-run and injection tests also show remaining harmful updates and condition-dependent degradation. Future work will focus on generation and inference efficiency for uncached utterances and recognition of rare emotion classes.

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A Implementation Details

A.1 Cache Construction and Calibration

Retrieval. TF–IDF fits lowercase unigrams and bigrams (at most 50,000 features) on training contexts from Section 3.3. Audio/visual cues are mean absolute value, SD, maximum absolute value, and mean absolute change from the preceding utterance. Standardize these cues with training means and population SD divisor $\max(\sigma_{\text{train}}, 10^{-6})$; L_2 normalize r^{AV} with divisor $\max(\|r^{\text{AV}}\|_2, 10^{-6})$. Validation/test reuse these statistics; retrieved contexts end at their examples.

Generation. The chat template adds a generation prompt. Greedy decoding uses `temperature=0`, `do_sample=false`, 4096 input and 160 output tokens, batch size 16, and bfloat16. Prompts include the speaker/utterance, causal context, five labeled references, and confusion guidance. Output is restricted to JSON fields `label`, `confidence`, `rationale`, `appraisal`, and `modality_hint`.

The anonymous artifact provides complete prompts, field ranges, JSON schema, parsing/default rules, calibration settings, and construction of five labeled causal-context examples.[1] Model inputs follow Equation 6.

Cost accounting. Wall time includes loading; loops cover retrieval, generation, parsing, and writing. One-thread Xeon Gold 5318Y retrieval averages 11.0772/13.1991 ms (256 queries, three rounds, 32 warm-ups), excluding feature extraction, LLM inference/tokenization, classification, networking, and writing.

Parsing and defaults. Parsing tries complete JSON, one brace-delimited extract, then flat recovery of label, confidence, and rationale, retaining raw responses without retries. Invalid labels or absent records give $M_i^{\text{cache}} = 0$ and zero input; absent rationales are empty. Missing/nonfinite appraisal or hints use Table 10; hints are nonnegative and renormalized. Missing/invalid raw confidence uses a deterministic fallback before calibration. The generic field fallback is $[0, 0, 0, \gamma_i]$ with uniform hints; a record still requires a valid task label to participate.

Table 10: Cache-field defaults.

Label	$[v, \alpha, d, \zeta]$	$[q^t, q^a, q^v]$
happy	[.70, .45, .35, .70]	[.45, .25, .30]
joy	[.85, .65, .45, .75]	[.40, .30, .30]
excited	[.85, .90, .50, .75]	[.25, .55, .20]
sad/sadness	[−.80, −.45, −.55, .65]	[.45, .35, .20]
neutral	[.00, .00, .00, .60]	[.55, .25, .20]
angry/anger	[−.75, .85, .65, .75]	[.25, .55, .20]
frustrated	[−.65, .70, −.35, .70]	[.35, .45, .20]
fear	[−.70, .80, −.70, .65]	[.35, .45, .20]
surprise	[.30, .85, .10, .55]	[.35, .25, .40]
disgust	[−.70, .55, .20, .70]	[.35, .25, .40]

Confidence calibration. Assign whole training dialogues to five folds $\mathcal{F}_f$ per dataset–generator pair. Parsed confidence is γ_i^{raw}; c_i^{in} follows field validation, defaults, and clipping to $[10^{-6}, 1 - 10^{-6}]$. Define $g_T(c) = \sigma(\text{logit}(c)/T)$, cached-label correctness $z_i^{\text{corr}} = \mathbb{I}(\hat{y}_i^{\text{llm}} = y_i)$, and binary cross-entropy BCE:

$$T^{(-f)} = \arg\min_{T>0} \sum_{i \in \mathcal{D}_{\text{train}} \setminus \mathcal{F}_f} \text{BCE}\Big(z_i^{\text{corr}}, g_T(c_i^{\text{in}})\Big). \tag{16}$$

Each record stores its split-specific confidence:

$$\gamma_i = \begin{cases} g_{T^{(-f)}}(c_i^{\text{in}}), & i \in \mathcal{F}_f, \\ \frac{1}{5}\sum_{f=1}^{5} g_{T^{(-f)}}(c_i^{\text{in}}), & i \in \text{Val} \cup \text{Test}. \end{cases} \tag{17}$$

Training forwards and diagnostics read the out-of-fold value; validation/test read mean output probabilities. Assignments are fixed before discriminative training; forwards read them directly while the trainable mixer recomputes ρ_i. Calibrators use training labels only; validation W-F1 selects thresholds and downstream hyperparameters.

Calibration metrics. Over Table 6's $N = 1{,}623/2{,}610$ valid caches, let $\mathcal{B}_b$ denote ten equal-width confidence bins and z_i^{corr} cached-label correctness:

$$\text{ECE} = N^{-1} \sum_{b=1}^{10} |\mathcal{B}_b| \, |\bar{z}_b - \bar{\gamma}_b|,$$
$$\text{Brier} = N^{-1} \sum_i (\gamma_i - z_i^{\text{corr}})^2. \tag{18}$$

Bins are $[0, .1), \ldots, [.9, 1]$; bars denote bin means, empty bins contribute zero, and ECE is in percent. Pre/post values are c_i^{in} and stored γ_i (Equation 17). Brier uses per-sample confidence; AUROC ranks cached-label correctness. Five positive-temperature maps and their mean preserve clipped-input order on test records.

A.2 Training Objectives

Affective cues. Parsed VAD defines

$$g_i^{\text{pos}} = \max(v_i, 0)\alpha_i, \quad g_i^{\text{low}} = \max(-v_i, 0)\max(-\alpha_i, 0),$$
$$g_i^{\text{ctrl}} = \max(-v_i, 0)\max(\alpha_i, 0) d_i. \tag{19}$$

Factor signs are retained; signed certainty is a separate field.

Branch losses. For mini-batch Ω and modality predictions $\hat{p}_i^m$, final cross-entropy is $\mathcal{L}_{\text{cls}} = -|\Omega|^{-1} \sum_{i \in \Omega} \log \hat{p}_{i,y_i}$. Dividing final/branch logits by $T_{\text{sd}} > 0$ gives $\widetilde{p}_i, \widetilde{p}_i^m$:

$$\mathcal{L}_{\text{uni}} = -\frac{1}{|\Omega|} \sum_{m \in \{t,a,v\}} \sum_{i \in \Omega} \log \hat{p}_{i,y_i}^m,$$
$$\mathcal{L}_{\text{sd}} = \frac{1}{|\Omega|} \sum_{m \in \{t,a,v\}} \sum_{i \in \Omega} \text{KL}(\text{sg}(\widetilde{p}_i) \| \widetilde{p}_i^m). \tag{20}$$

sg denotes stop-gradient. Both datasets use $T_{\text{sd}} = 1/8$, without a T_{sd}^2 multiplier.

LLM-driven and source-weight alignment. Define $\bar{w}_i^{\text{llm}} = \text{sg}(A_i \rho_i \gamma_i)$ and $\bar{w}_i^{\text{rel}} = \text{sg}(A_i \, \text{clip}(\rho_i \gamma_i, 0, 1))$. LLM alignment is

$$\mathcal{L}_{\text{llm}} = \frac{\sum_{i \in \Omega} \bar{w}_i^{\text{llm}} \, \text{KL}(p_i^{\text{llm}} \| \hat{p}_i)}{\max(\sum_{i \in \Omega} \bar{w}_i^{\text{llm}}, 1)}. \tag{21}$$

Equation 5 defines p_i^{llm}. Source-weight supervision uses the detached target $\tilde{\beta}_i^* = \text{sg}(\beta_i^*)$:

$$\mathcal{L}_{\text{rel}} = \frac{\sum_{i \in \Omega} \bar{w}_i^{\text{rel}} \|\beta_i - \tilde{\beta}_i^*\|_2^2}{\max(\sum_{i \in \Omega} \bar{w}_i^{\text{rel}}, 1)}. \tag{22}$$

[1] Anonymous implementation artifact

For a vector x, define $\mathcal{N}(x) = \max(x, 10^{-6})/\sum_j \max(x_j, 10^{-6})$ element-wise and $[x]_+ = \max(x, 0)$. Cue scores ℓ_i^m, modality target m_i, appraisal strength χ_i, and cognitive share π_i^c are

$$\begin{aligned}
\ell_i^t &= |\zeta_i| + |g_i^{\text{low}}| + [1 - |\alpha_i|]_+, \\
\ell_i^a &= |\alpha_i| + |g_i^{\text{ctrl}}|, \quad \ell_i^v = [v_i]_+ + |g_i^{\text{pos}}| + |d_i|, \\
m_i &= \mathcal{N}(q_i + \mathcal{N}([\ell_i^t, \ell_i^a, \ell_i^v])), \\
\chi_i &= \text{clip}\left(\frac{|v_i| + |\alpha_i| + |d_i| + \|g_i\|_1}{3}, 0, 1\right), \\
\pi_i^c &= \gamma_i \rho_i \chi_i, \quad \beta_i^* = [(1 - \pi_i^c) m_i; \pi_i^c].
\end{aligned} \tag{23}$$

Weights, normalizers, and targets are recomputed each forward pass; prediction-path ρ_i remains differentiable.

Affective geometry. Normalize final representations and $[v_i, \alpha_i, d_i]$ by L_2 (guard 10^{-12}), giving $\bar{h}_i, \bar{\mathbf{a}}_i^{\text{VAD}}$; exclude ζ_i and retain zero VAD. For admitted set $\mathcal{I} = \{i \in \Omega : A_i = 1\}$ and ordered pairs $\mathcal{P} = \{(i, j) : i, j \in \mathcal{I}, i \neq j, y_i = y_j\}$, use $T_{\text{aff}} > 0$ to define

$$s_{ij} = \bar{h}_i^\top \bar{h}_j / T_{\text{aff}}, \qquad \kappa_{ij} = \exp(-\|\bar{\mathbf{a}}_i^{\text{VAD}} - \bar{\mathbf{a}}_j^{\text{VAD}}\|_2). \tag{24}$$

The same-class loss is

$$\mathcal{L}_{\text{pos}} = -\frac{\sum_{(i,j)\in\mathcal{P}} \kappa_{ij} \log \frac{\exp(s_{ij})}{\sum_{k\in\mathcal{I}\setminus\{i\}} \exp(s_{ik})}}{\max(|\mathcal{P}|, 1)}. \tag{25}$$

For admitted ordered pairs $\mathcal{N}_{\text{pair}}$ whose labels belong to training-derived C_{conf} in either order, use margin $m_{\text{aff}} \geq 0$ and $\mathcal{L}_{\text{aff}} = \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{neg}}$:

$$\mathcal{L}_{\text{neg}} = \frac{\sum_{(i,j)\in\mathcal{N}_{\text{pair}}} \kappa_{ij} [\bar{h}_i^\top \bar{h}_j - m_{\text{aff}}]_+}{\max(|\mathcal{N}_{\text{pair}}|, 1)}. \tag{26}$$

Empty pair sets contribute zero. Before ECRA training, first-seed five-fold out-of-fold Base predictions on training dialogues determine C_{conf}. Using confusion counts M_{ab}, class counts n_a, and fixed VAD prototypes v_a, rank pairs by $(M_{ab} + M_{ba})/(n_a + n_b) + \exp(-\|v_a - v_b\|_2)$. The top three (ties by label index) are happy–excited, angry–frustrated, neutral–sad on IEMOCAP; fear–surprise, anger–disgust, neutral–sadness on MELD.

Both datasets set $\lambda_{\text{uni}} = \lambda_{\text{sd}} = 1$, $\lambda_{\text{rel}} = 10^{-1}$, and $T_{\text{aff}} = m_{\text{aff}} = 0.2$. IEMOCAP/MELD use $\lambda_{\text{llm}} = 5 \times 10^{-1}/10^{-1}$ and $\lambda_{\text{aff}} = 2 \times 10^{-1}/3 \times 10^{-1}$.

B Additional Experiments and Analyses

B.1 Fusion Controls and Gate Variants

Baseline inputs. Method-specific inputs and implementation details are documented in the anonymous artifact referenced in Appendix A.1.

Base and Residual Fusion Backbone. Base / w/o cache uses Equation 3 and $\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{uni}} + \mathcal{L}_{\text{sd}}$, without cognitive inputs or losses. RFB retains μ_i and uses $\mathbf{u}_i^{\text{RFB}} = \text{GELU}(W_1[e_i^t; e_i^a; e_i^v] + b_1)$, a width-12 bottleneck:

$$\begin{aligned}
\beta_i^{\text{RFB}} &= \text{softmax}(W_\beta \mathbf{u}_i^{\text{RFB}} + b_\beta), \\
\eta_i^{\text{RFB}} &= \sigma((w_\eta^{\text{RFB}})^\top \mathbf{u}_i^{\text{RFB}} + b_\eta^{\text{RFB}}),
\end{aligned} \tag{27}$$

$$h_i^{\text{RFB}} = h_i^{\text{base}} + \eta_i^{\text{RFB}} \left(\sum_{m\in\{t,a,v\}} \beta_i^{\text{RFB},m} e_i^m - h_i^{\text{base}} \right). \tag{28}$$

Table 11: Fusion controls and native readouts.

Setting	IEMOCAP		MELD	
	ACC	W-F1	ACC	W-F1
Base	70.52 ± 0.26	70.76 ± 0.11	65.89 ± 0.36	64.43 ± 0.08
Base + prob.	71.71 ± 0.15	71.93 ± 0.23	66.68 ± 0.17	65.36 ± 0.06
ECRA-base	71.18 ± 0.15	71.36 ± 0.16	66.28 ± 0.16	65.40 ± 0.25
ECRA-base + prob.	71.81 ± 0.16	71.98 ± 0.16	66.85 ± 0.14	65.99 ± 0.16
ECRA	74.71 ± 0.13	74.87 ± 0.12	68.34 ± 0.15	67.51 ± 0.12

With $W_1 \in \mathbb{R}^{12\times3072}$, $W_\beta \in \mathbb{R}^{3\times12}$, $w_\eta^{\text{RFB}} \in \mathbb{R}^{12}$, biases, and a reused classifier, RFB adds 36,928 parameters. Initialize $w_\eta^{\text{RFB}} = 0$, $b_\eta^{\text{RFB}} = \text{logit}(.05) \approx -2.944439$; shared parameters match per seed, with fixed linear-layer initialization elsewhere.

RFB shares Base's objectives and uses its own stopped-gradient final predictions; other settings follow Section 4.1.

Retrieval-derived structured cache. Non-LLM Cache retains downstream architecture/objectives. Similarity-normalized labels give p_i^{ret} and weight fixed class VAD prototypes; normalized modality similarities give q_i^{ret}, with $\gamma_i^{\text{ret}} = \max_k p_{i,k}^{\text{ret}}$. Support uses the similarities and agreement against $\arg\max_k p_{i,k}^{\text{ret}}$.

Trained fusion controls. Base + prob., ECRA-base + prob., and ECRA-base + logit train independently from scratch with fusion active during training and evaluation (Section 4.1). The former uses Base; the latter two use ECRA-family architectures. ECRA-base and ECRA read native ECRA's backbone/fused paths (Table 11). For logits $z_i^{\text{base}} = W_o h_i^{\text{base}} + b_o$, define:

$$\omega_i = \kappa A_i \tau_i,$$

$$p_i^{\text{prob}} = (1 - \omega_i)\hat{p}_i^{\text{base}} + \omega_i p_i^{\text{llm}}, \tag{29}$$

$$p_i^{\text{logit}} = \text{softmax}\left((1 - \omega_i) z_i^{\text{base}} + \omega_i \log \bar{p}_i^{\text{llm}}\right). \tag{30}$$

$\bar{p}_i^{\text{llm}} = \mathcal{N}(p_i^{\text{llm}})$ floors priors at 10^{-6} and renormalizes. Unavailable/rejected caches return backbone predictions before the logarithm. Per paired seed, the frozen ECRA support mixer supplies common ρ_i, A_i, τ_i at $\delta_q = 0.60$, $\delta_c = 0.85$. Select $\kappa \in \{0, 0.05, \ldots, 1.00\}$ per seed/operator by mean validation W-F1 of the trained Base and ECRA-family variants; ties favor smaller values. Both probability-fusion variants share κ and these fixed inputs. Evaluation disables dropout. No learned fusion head is added; SDs cover five training runs.

E2E-ηGate. E2E-ηGate retains ECRA's four-source weights β_i and h_i^{rel} in Equation 11, replacing the outer coefficient:

$$\begin{aligned}
x_i^{\text{gate}} &= [h_i^{\text{base}}; h_i^c; \rho_i; \gamma_i; q_i; \mathbf{a}_i; g_i], \\
a_i^{\text{gate}} &= \sigma(\text{MLP}(x_i^{\text{gate}})), \\
h_i^{\text{final}} &= h_i^{\text{base}} + A_i a_i^{\text{gate}} (h_i^{\text{rel}} - h_i^{\text{base}}).
\end{aligned} \tag{31}$$

The $2060 \to 32 \to 1$ gate uses GELU, dropout 0, zero output weights, and bias $\text{logit}(0.05) = -2.944438979$, initially giving accepted/rejected coefficients 0.05/0. This variant follows Full ECRA's data, features, cache, optimization, other losses, and validation selection (Section 4.1), with shared initializations paired by seed. Its separately trained support mixer computes its own admission. Equation 22 directly supervises fusion weights β_i with the same target, normalization, gradient handling, and $\lambda_{\text{rel}} = 10^{-1}$.

Table 12: Cache-path, prompt, and gate controls.

Setting	IEMOCAP		MELD	
	ACC	W-F1	ACC	W-F1
w/o class-prior path	72.63 ± 0.15	72.91 ± 0.09	67.32 ± 0.23	65.97 ± 0.16
w/o VAD path	73.09 ± 0.21	73.28 ± 0.14	67.68 ± 0.17	66.69 ± 0.08
w/o q	72.86 ± 0.14	73.12 ± 0.11	67.49 ± 0.11	66.14 ± 0.15
w/o a^{label}	73.75 ± 0.11	73.93 ± 0.12	67.63 ± 0.12	66.46 ± 0.15
w/o confusion prompt	74.43 ± 0.14	74.58 ± 0.12	67.91 ± 0.07	67.28 ± 0.15
Soft gate	72.64 ± 0.08	72.78 ± 0.15	66.97 ± 0.16	66.35 ± 0.40
Straight-through	72.69 ± 0.15	72.84 ± 0.13	67.23 ± 0.19	66.42 ± 0.17
Continuous trust	73.83 ± 0.25	74.03 ± 0.15	67.67 ± 0.15	66.60 ± 0.24
Full ECRA	**74.71 ± 0.13**	**74.87 ± 0.12**	**68.34 ± 0.15**	**67.51 ± 0.12**

Table 13: Main-run correction and harm.

Dataset	Method	B	H	Correct base/final	H_{err}
IEMOCAP	ECRA	84	20	1,151/1,215	6/82 (7.32%)
	E2E Cache Gate	65	49	1,180/1,196	20/86 (23.26%)
MELD	ECRA	93	27	1,724/1,790	9/121 (7.44%)
	E2E Cache Gate	65	44	1,750/1,771	25/125 (20.00%)

Table 14: Wrong-label cache injection.

p	IEMOCAP		MELD	
	ECRA	E2E	ECRA	E2E
0	74.99	73.82	67.62	66.87
0.1	74.30 ± 0.20	72.45 ± 0.24	67.03 ± 0.24	66.84 ± 0.08
0.3	72.88 ± 0.38	69.61 ± 0.35	66.94 ± 0.16	64.65 ± 0.20
0.5	71.28 ± 0.45	66.93 ± 0.34	65.91 ± 0.25	62.60 ± 0.14

Shared-cache control. The earlier E2E Cache Gate used in the main-run diagnostics and Figure 3 retains a four-source weight head trained by $\mathcal{L}_{\text{rel}}$; its prediction path uses the gated additive residual:

$$h_i^{\text{final}} = h_i^{\text{base}} + A_i a_i^{\text{gate}} \tanh(W_\Delta [h_i^{\text{base}}; h_i^c]). \tag{32}$$

W_Δ maps to d_h dimensions.

Supplementary controls. In Table 12, *w/o confusion prompt* regenerates caches without the confusion-guidance sentence, retaining the training confusable-pair set and pairwise loss. *w/o VAD path* removes affective appraisal, derived guidance, and $\mathcal{L}_{\text{aff}}$; *w/o class-prior path* removes the p^{llm} input and $\mathcal{L}_{\text{llm}}$. The latter two assess input–supervision pathways. Removing q removes hints; removing a^{label} retains the three similarities.

Gate variants. Soft uses a validation-selected temperature $T_g > 0$:

$$G_i^{\text{soft}} = M_i^{\text{cache}} \sigma((\rho_i - \delta_q)/T_g) \sigma((\gamma_i - \delta_c)/T_g). \tag{33}$$

Straight-through uses $\text{sg}(A_i - G_i^{\text{soft}}) + G_i^{\text{soft}}$; continuous trust uses $M_i^{\text{cache}} \rho_i \gamma_i$. For gate G_i and $\bar{\beta}_i = \text{softmax}(o_i + b_i)$, normalize with $D_i = \sum_{m \in \{t,a,v\}} \bar{\beta}_i^m + G_i \bar{\beta}_i^c$: $\beta_i^c = G_i \bar{\beta}_i^c / D_i$, $\beta_i^m = \bar{\beta}_i^m / D_i$. Residual correction and the scalar alignment-loss weights use G_i in place of A_i; binary gates recover the hard mask.

Straight-through retains the binary forward rule and changes backward gradients.

B.2 Admission and Correction Diagnostics

Gate optimization and stability. For records with $A_i = 0$, cache-dependent auxiliary terms are disabled, while classification and branch losses remain active. Re-entry requires valid caches meeting δ_c and support crossing δ_q. Epoch-wise evaluation disables dropout/updates and uses out-of-fold confidence (Equation 17). Admission rates use all training records as the denominator. At validation-selected checkpoints, IEMOCAP/MELD training admission is 38.43 ± 2.13%/33.88 ± 1.41%; ever-rejected re-entry (any later acceptance by the selected checkpoint, including records rejected again afterward) is 5.35 ± 1.38%/4.27 ± 1.66%. Final-admission Jaccard (intersection over union) over ten distinct seed pairs is 0.86 ± 0.03/0.82 ± 0.04 (minima .81/.75). SDs describe training variability.

Correction and harm. For the main-run counts in Table 13, let $E = \{i : \hat{y}_i^{\text{llm}} \neq y_i,\ \gamma_i \geq 0.85\}$:

$$H_{\text{err}} = \frac{\#\{i \in E : \hat{y}_i^{\text{base}} = y_i,\ \hat{y}_i^{\text{final}} \neq y_i\}}{\#\{i \in E : \hat{y}_i^{\text{base}} = y_i\}}. \tag{34}$$

Main-run ACC gain over each system's same-forward backbone is $100(B - H)/N$ points. ECRA's backbone/final ACC is 70.92/74.86% and 66.05/68.58%, gaining 3.94/2.53 points.

Main-run admission retains 500 correct and 124 incorrect cached labels on IEMOCAP, and 691/193 on MELD (Table 6).

Matched-budget diagnostic. Rank valid caches by unrounded γ_i (confidence only) or $s_i^{\text{joint}} = \min(\gamma_i/0.85, \rho_i/0.60)$ (confidence + evidence). Keep 487/783 per checkpoint (IEMOCAP/MELD), breaking ties by ascending dialogue ID, then utterance index; no test labels are used. Section 4.5 specifies frozen modules and outer-gate fitting; main runs retain Equation 9.

Net summarizes per-checkpoint $B - H$. In Table 7's row order, means ± sample SD are IEMOCAP: 35.2±0.84, 44.0±2.35, 43.2±2.39, 53.0 ± 2.35; MELD: 39.6 ± 2.07, 48.4 ± 2.30, 49.4 ± 3.21, 58.4 ± 3.65.

B.3 Robustness and Qualitative Analysis

Wrong-label cache injection. Figure 3 uses seeds 2026–2030 for shared perturbed records and wrong labels, with post-calibration confidence 0.95. Class priors, retrieved-label agreement, and frozen-mixer support are recomputed; other fields stay fixed. Systems compute their own admission. Table 14 gives W-F1 (%); E2E denotes E2E Cache Gate. Each column uses one checkpoint: $p = 0$ is clean; $p > 0$ gives mean ± sample SD over five perturbations. Table 8's shared-proposal experiment instead fixes clean admitted IDs.

Modality masking and qualitative analysis. With caches and encodings fixed, branch masking renormalizes fusion weights over seven availability patterns. Removing text yields W-F1 of 63.44%/39.93% on IEMOCAP/MELD; text-only fusion yields 66.56%/65.30%.

Class-mean appraisals place happy/excited/joy at positive valence, sadness at low arousal, and angry/frustrated at high arousal.

For two admitted traces, IEMOCAP's audio weight is .37 (angry/frustrated probabilities .64/.22); MELD's text weight is .43 (neutral .71). Equation 12 gives cognitive/backbone coefficients $\eta_i \beta_i^c / 1 - \eta_i$: .055/.79 and .035/.84, respectively.